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1. Introduction

This course will be about Sen theory, which is a subset of p-adic Hodge theory, which is a p-adic analogue of complex Hodge theory. For a compact Riemann surface $X/\mathbb{C}$, there is a Hodge decomposition

$$H^1(X, \mathbb{C}) = H^0(X, \Omega^1_X) \oplus H^1(X, \mathcal{O}_X),$$

and there is a symmetry coming from the complex conjugation, namely the nontrivial element of $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$.

Tate discovered a p-adic analogue as follows. For a projective smooth curve X over $\mathbb{Q}_p$[1], we have an isomorphism of $G_{\mathbb{Q}_p}$-modules,

$$H^1_{\text{ét}}(X_{\mathbb{C}_p}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} \mathbb{C}_p = H^0(X_{\mathbb{C}_p}, \Omega^1)(-1) \oplus H^1(X_{\mathbb{C}_p}, \mathcal{O}),$$

where $\mathbb{C}_p$ is the p-adic completion of $\overline{\mathbb{Q}}_p$, and (-1) means the inverse Tate twist (inverse of cyclotomic character). This is called the Hodge–Tate decomposition.

Sen had discovered a much more general decomposition. For a continuous finite-dimensional $G_{\mathbb{Q}_p}$-representation $V/\mathbb{Q}_p$, there is a decomposition of $V \otimes_{\mathbb{Q}_p} \mathbb{C}_p$. Namely, there is a **Sen operator** $\theta_V \in \mathrm{End}_{\mathbb{C}_p}(V \otimes_{\mathbb{Q}_p} \mathbb{C}_p)$ which commutes with $G_{\mathbb{Q}_p}$-action. Thus there is an eigenspace decomposition

$$W := V \otimes_{\mathbb{Q}_p} \mathbb{C}_p = \bigoplus_{\lambda \text{ eigenvalues of } \theta_V} W_\lambda.$$

When $V = H^1_{\text{ét}}(X_{\mathbb{C}_p}, \mathbb{Q}_p)$, this recovers the Hodge–Tate decomposition. The set $\{-\lambda\}$ is called the set of generalized Hodge–Tate weights of V.

[1] Or over a finite extension of $\mathbb{Q}_p$. Now we know this holds even over $X/\mathbb{C}_p$ but this was not known at the times of Tate.

How do we construct θ_V? There are two steps:

(1) (Almost étale descent) descend from $\mathbb{C}_p$ to $\widehat{\mathbb{Q}_p(\mu_{p^\infty})}$;

(2) (Decompletion) descend from $\widehat{\mathbb{Q}_p(\mu_{p^\infty})}$ to $\mathbb{Q}_p(\mu_{p^\infty})$.

We will spend some time on this. Then some more topics:

- "universal Sen operator" θ_{Sen}, in something like "$(\mathbb{C}_p \otimes_{\mathbb{Q}_p} \operatorname{Lie} G_{\mathbb{Q}_p})^{G_{\mathbb{Q}_p}}$";
- The work of Berger–Colmez on p-adic Cauchy–Riemann operator;
- The work of Fontaine on Sen theory over B_{dR};
- The work of Faltings and others on relative Sen theory, or the "p-adic Simpson correspondence" (contingent upon the uniformization..).

2. $\mathbb{C}_p$

The field $\mathbb{C}_p$ has a norm $|\cdot| : \mathbb{C}_p \to \mathbb{R}_{\geq 0}$, $p \mapsto p^{-1}$, and a valuation $v : \mathbb{C}_p \to \mathbb{R} \cup \{+\infty\}$, given by $v = -\log_p |\cdot|$.

Theorem 2.1. *$\mathbb{C}_p$ is algebraically closed. More generally, for a complete (?) non-archimedean field K, then $\widehat{\overline{K}}$ is algebraically closed.*

This is due to Krasner's lemma.

Lemma 2.2. *Let K be a complete nonarchimedean field, and let $\alpha, \beta \in K^s$. If $|\alpha - \beta| < |\alpha - \alpha'|$ for any conjugate α' of α, then $\alpha \in K(\beta)$.*

Proof. Let $\gamma = \alpha - \beta$. This is in $K(\alpha, \beta)$, which is a separable extension of $K(\beta)$. Let γ' be a conjugate of γ over $K(\beta)$. It is of form $\gamma' = \alpha' - \beta$ for a conjugate α' of α. Note that $|\gamma| = |\gamma'|$, so $|\alpha - \beta| = |\alpha' - \beta|$. By the triangle inequality, this is a contradiction if $\alpha \neq \alpha'$, so $\gamma' = \gamma$. This shows that $\gamma \in K(\beta)$, or $\alpha \in K(\beta)$. $\square$

Proof that $\mathbb{C}_p$ is algebraically closed. Suppose we have $P(X) = X^d + \sum_{i=0}^{d-1} a_i X^i \in \mathbb{C}_p[X]$. Let α be a root of $P(X)$ in $\overline{\mathbb{C}_p}$. Let $\epsilon = \min_{\alpha \neq \alpha' \text{ conjugate over } \mathbb{C}_p} |\alpha - \alpha'|$. Choose a monic $Q(X) \in \overline{\mathbb{Q}}_p[X]$ approximating $P(X)$ well enough so that $|Q(\alpha)| < \epsilon^d$. Then, if we denote β_i's to be the roots of $Q(X)$, then there is i such that $|\alpha - \beta_i| < \epsilon$. By Krasner's lemma, $\alpha \in \mathbb{C}_p(\beta_i) = \mathbb{C}_p$. $\square$

For a closed subgroup $H \leq G_{\mathbb{Q}_p}$, what is the relation between $\mathbb{C}_p^H$ and $\overline{\mathbb{Q}}_p^H$?

Theorem 2.3. *$\mathbb{C}_p^H = \widehat{\overline{\mathbb{Q}}_p^H}$. In particular, for a finite extension $K/\mathbb{Q}_p$, $\mathbb{C}_p^{G_K} = K$.*

Proof. To prove this, we need a lemma by Ax and Sen.

Definition 2.4. For a complete nonarchimedean field K and E/K algebraic, for $\alpha \in K^s$, let

$$\Delta_E(\alpha) = \min_{\alpha' \text{ conjugates of } \alpha \text{ over } E} \min\{v(\alpha - \alpha')\}.$$

Lemma 2.5 (Ax–Sen)**.** *For a complete nonarchimedean field K and E/K algebraic, for $\alpha \in K^s$, there is $a \in E$ such that $v(\alpha - a) > \Delta_E(\alpha) - \frac{p}{(p-1)^2}$.*

Assuming this, we prove the theorem. Let $\alpha \in \mathbb{C}_p^H$ and $L = \overline{\mathbb{Q}}_p^H$. We can choose $\alpha_n \in \overline{\mathbb{Q}}_p$ that approximates α so that $v(\alpha - \alpha_n) > n$. For all $g \in H$,

$$v(g\alpha_n - \alpha_n) = v(g(\alpha_n - \alpha) + (\alpha - \alpha_n)) > n.$$

This means that $\Delta_L(\alpha_n) > n$.

By Ax–Sen with $E = L$, $K = \mathbb{Q}_p$ and $\alpha = \alpha_n$, there is $a_n \in L$ such that $|\alpha_n - a_n| > n - \frac{p}{(p-1)^2}$. This means a_n also approaches α. $\square$

Proof of Ax–Sen lemma. We assume for simplicity that $\operatorname{char} K = 0$. The idea is to induct on the degree of α over E, denoted $d = d(\alpha, E)$.

Lemma 2.6. *Let $R(X) \in K^s[X]$ be a degree $d \geq 2$ monic polynomial such that $v(\lambda) \geq \gamma$ for all roots λ of $R(X)$. For $0 < m < d$, there is a root μ of $R^{(m)}(X)$ such that $v(\mu) \geq \gamma - \frac{1}{d-m} v(\binom{d}{m})$.*

Proof. Let $P(X) = \prod_{i=1}^{d}(X - \lambda_i) = X^d + \sum_{i=0}^{d-1} b_i X^i$. Then, b_m is an integral polynomial of $\lambda_1, \cdots, \lambda_d$ of degree $d - m$. Thus, $v(b_m) \geq (d-m)\gamma$. Let

$$\frac{1}{m!} R^{(m)}(X) = \binom{d}{m}(X - \mu_1) \cdots (X - \mu_{d-m}).$$

Then, we have

$$b_m = \binom{d}{m}(-1)^{d-m} \mu_1 \cdots \mu_{d-m}.$$

Thus,

$$v(\mu_1) + \cdots + v(\mu_{d-m}) \geq (d-m)r - v(\binom{d}{m}).$$

Thus, there is $1 \leq i \leq d - m$ such that $v(\mu_{d-m})$ is $\geq$ the average of RHS. $\square$

To use the lemma well, we will induct on the formula

$$v(a - \alpha) > \Delta_E(\alpha) - \sum_{i=1}^{\lfloor \log_p d \rfloor} \frac{1}{p^i - p^{i-1}}.$$

Let $R(X) = P(X + \alpha)$, where $P(X)$ is the monic characteristic polynomial of α over E. Then $v(\lambda) \geq \gamma$ for all roots γ of $R(X)$, where $\gamma = \Delta_E(\alpha)$. Let $m = p^s$, where $d = p^s n$ is a decomposition where $n = p$ if $d = p^N$ for $N \geq 1$, and $(n, p) = 1$ otherwise. Then

$$v_p(\binom{d}{m}) = \begin{cases} 0 & \text{if } (n,p) = 1 \\ 1 & \text{if } n = p. \end{cases}$$

This is to obtain the smallest possible $v_p(\binom{d}{m})$. By the lemma, there is a root μ of $R^{(m)}(X)$ such that $v(\mu) \geq \gamma - \frac{1}{d-m} v(\binom{d}{m})$. Let $\mu = \beta - \alpha$. Then $d(\beta, E) \leq d - m < d$. If $\beta \in E$, we are done with the given inequality. If not, we use the induction hypothesis. $\square$

3. Almost étale descent

Suppose there is a G-covering $\pi : X \to Y$ of topological spaces, where G is a finite group. Let $\mathcal{F}$ be a sheaf on Y, and $\mathcal{G} = \pi^* \mathcal{F}$. Then $\mathcal{F}$ can be recovered from $\mathcal{G}$ plus descent datum, namely $\mathcal{G} \cong g^* \mathcal{G}$ plus cocycle condition. In the algebraic geometry context, there is étale descent.

Example 3.1 (Galois descent)**.** Let E/F be a finite Galois extension, $G = \operatorname{Gal}(E/F)$ and V be a finite dimensional vector space over E, equipped with a semilinear action of G on V, namely

$$g(av) = g(a)g(v), \quad a \in E, v \in V.$$

This semilinear action is a descent datum, and this is effective, in that $\dim_F V^G = \dim_E V$ and $V^G \otimes_F E \xrightarrow{\sim} V$.

We sketch the proof of this. In the case $V = E$, it is quite trivial (although it uses the full force of Galois theory). We furthermore have

$$E \otimes_F E \xrightarrow{(a,b)\mapsto(g(a)b)_{g\in G}} \bigoplus_G E = C(G, E),$$

and this isomorphism is G-equivariant, in that the left hand side has the G-action on the first factor, and the right hand side has G-action coming from right translation on G.

Replacing E by V, we have

$$V \otimes_F E = V \otimes_E (E \otimes_F E) \cong V \otimes_E \bigoplus_G E = \bigoplus_{g\in G} V \otimes_{E,g} E.$$

Here the G-action on the rightmost part is the right-translation on the indices. Take the embedding $V \hookrightarrow \oplus_{g\in G} V \otimes_{E,g} E$ where v is sent to $(v, 0, \cdots, 0)$ where the nonzero index corresponds to the identity element, we have the rightmost side isomorphic as an G-module to $E \otimes_F V_0$ where the G-action comes from the G-action on E. This shows that $\dim_F V^G = \dim_E V$.

Note that showing that $E \otimes_F E \xrightarrow{\sim} \bigoplus_G E$ is equivalent to showing that the matrix is invertible, where the transition matrix can be given by

$$\begin{pmatrix} \sigma_1 e_1 & \sigma_1 e_2 & \cdots \\ \sigma_2 e_1 & \sigma_2 e_2 & \cdots \\ \cdots & \cdots & \cdots \end{pmatrix},$$

for an F-basis $\{e_1, \cdots, e_n\}$ of E. Note that the matrix for the trace pairing $E \times E \to F$ with the given basis is $B^T B$, so showing that the map is an isomorphism is equivalent to showing that the trace pairing is perfect, or $1 \in \mathrm{tr}(E)$, or $\mathrm{tr}(E) = F$.

The same strategy works when we replace E, F by $\mathcal{O}_E, \mathcal{O}_F$ for an unramified extension of local fields E/F, namely $\mathrm{tr}(\mathcal{O}_E) = \mathcal{O}_F$. So there is an étale descent.

What is almost étale descent? Tate discovered that something happens when one does a highly ramified field extension. Consider $\mathbb{Q}_p(\mu_{p^\infty}) \supset \mathbb{Z}_p[\mu_{p^\infty}]$. The maximal ideal $\mathfrak{m}_\infty$ is non-Noetherian (as $p^{\frac{k}{(p-1)p^{n-1}}} \in \mathfrak{m}_\infty$ for all $n, k > 0$, as $v(\zeta_{p^n} - 1) = \frac{1}{(p-1)p^{n-1}}$). Still, Tate discovered the following

Theorem 3.2 (Tate)**.** *For a finite Galois extension L of $\mathbb{Q}_p(\mu_{p^\infty})$, $\mathrm{tr}(\mathcal{O}_L) \supset \mathfrak{m}_\infty$. Equivalently, $\mathbb{Z}_p[\mu_{p^\infty}]/\mathrm{tr}(\mathcal{O}_L)$ is killed by any p^ϵ, $\epsilon > 0$. Equivalently, $\Omega_{\mathcal{O}_L/\mathbb{Z}_p[\mu_{p^\infty}]}$ is killed by p^ϵ for any $\epsilon > 0$.*

This is referred as that $\mathcal{O}_L$ is **almost étale** over $\mathbb{Z}_p[\mu_{p^\infty}]$.

Definition 3.3. For a subfield $K \subset \overline{\mathbb{Q}}_p$, assume that $\mathfrak{m}_K \otimes_{\mathcal{O}_K} \mathfrak{m}_K = \mathfrak{m}_K$. Equivalently, assume that there is a sequence of elements in $\mathfrak{m}_K$ whose valuation goes to zero. For a finite extension L/K, $\mathcal{O}_L/\mathcal{O}_K$ is **almost étale** if $\mathrm{tr}_{L/K} \mathcal{O}_L \supset \mathfrak{m}_K$. We say $\overline{\mathbb{Z}}_p/\mathcal{O}_K$ is almost étale if $\mathcal{O}_L/\mathcal{O}_K$ is almost étale for any finite extension L/K.

Tate's theorem implies that, for any algebraic extension $K/\mathbb{Q}_p(\mu_{p^\infty})$, $\overline{\mathbb{Z}}_p$ is almost étale over $\mathcal{O}_K$.

This is useful in several applications. Suppose L/K is a finite Galois extension with $\mathcal{O}_L/\mathcal{O}_K$ almost étale. Let $G = \mathrm{Gal}(L/K)$.

(1) **Galois cohomology.** For an $\mathcal{O}_L$-module M with a semilinear action of G, $\widehat{H}^i(G, M)$, as an $\mathcal{O}_K$-module, is almost zero (i.e. annihilated by $\mathfrak{m}_K$). Here, $\widehat{H}$ is the Tate cohomology.

The proof is simple. The $\mathcal{O}_L$-action on M gives a map $\mathcal{O}_L \times M \to M$, giving rise to a cup product $\widehat{H}^i(G, \mathcal{O}_L) \times \widehat{H}^j(G, M) \to \widehat{H}^{i+j}(G, M)$. Note however that the $\mathcal{O}_K$-action on $\widehat{H}^i(G, M)$ is really the action of $H^0(G, \mathcal{O}_L)$ via the cup product, and the formalism shows that this action factors through the quotient $\widehat{H}^0(G, \mathcal{O}_L) = \mathcal{O}_K / \operatorname{tr} \mathcal{O}_L$, which is almost zero. This shows that $\mathfrak{m}_K \widehat{H}^i(G, M) = 0$.

For example, if one applies this to $M = \mathcal{O}_L/p^n$, then this shows that $H^0(G, \mathcal{O}_L/p^n) \overset{\mathsf{a}}{=} \operatorname{tr}(\mathcal{O}_L/p^n) \overset{\mathsf{a}}{=} \mathcal{O}_K/p^n$. If $\overline{\mathbb{Z}}_p/\mathcal{O}_K$ is almost étale, taking direct limits over L, we get

$$H^0(G_K, \overline{\mathbb{Z}}_p/p^n) \overset{\mathsf{a}}{=} \mathcal{O}_K/p^n.$$

By taking the inverse limit on n, we have

$$H^0(G_K, \mathcal{O}_{\mathbb{C}_p}) \overset{\mathsf{a}}{=} \mathcal{O}_{\widehat{K}}.$$

Since both sides are non-p-torsion, by inverting p, we get $H^0(G_K, \mathbb{C}_p) = \widehat{K}$, which is what we proved before.

(2) **Almost étale descent.** For an $\mathcal{O}_L$-module M with a semilinear G-action, we have

$$M^G \otimes_{\mathcal{O}_K} \mathcal{O}_L \overset{\mathsf{a}}{=} M.$$

This is because

$$\mathcal{O}_L \otimes_{\mathcal{O}_K} \mathcal{O}_L \overset{\mathsf{a}}{=} \bigoplus_G \mathcal{O}_L,$$

by the same reason, $\operatorname{tr}(\mathcal{O}_L) \overset{\mathsf{a}}{=} \mathcal{O}_K$! For an infinite extension story, if $\overline{\mathbb{Z}}_p/\mathcal{O}_K$ is almost étale, the you would want a discrete $\overline{\mathbb{Z}}_p$-module (or a discrete $\mathcal{O}_{\mathbb{C}_p}$-module) M with a semilinear G_K-action. Here, discrete means that $M[p^\infty] = M$ and each vector is fixed by a finite index subgroup of G_K. Then

$$M^{G_K} \otimes_{\mathcal{O}_{\widehat{K}}} \mathcal{O}_{\mathbb{C}_p} = M^{G_K} \otimes_{\mathcal{O}_K} \overline{\mathbb{Z}}_p \overset{\mathsf{a}}{=} M.$$

In particular, for a finite-dimensional G_K-representation V over $\mathbb{Q}_p$, consider $W = V \otimes_{\mathbb{Q}_p} \mathbb{C}_p$, equiipped with a diagonal G_K-action, regarded as a semilinear G_K-module over $\mathbb{C}_p$. For a G_K-invariant lattice $T \subset V$, let $M = T \otimes_{\mathbb{Z}_p} \mathcal{O}_{\mathbb{C}_p}$ and $M_n = T/p^n$. Then, we know that $H^{>0}(G_K, M_n) \overset{\mathsf{a}}{=} 0$ and $M_n^{G_K} \otimes_{\mathcal{O}_{\widehat{K}}} \mathcal{O}_{\mathbb{C}_p} \overset{\mathsf{a}}{=} M_n$. From this, one can easily prove that

$$M^{G_K} \otimes_{\mathcal{O}_{\widehat{K}}} \mathcal{O}_{\mathbb{C}_p} \overset{\mathsf{a}}{=} M.$$

By inverting p, we get

$$W^{G_K} \otimes_{\widehat{K}} \mathbb{C}_p = W.$$

Proof of Tate's theorem, Theorem 3.2. Let $\mathbb{Q}_p(\mu_{p^\infty})$ be denoted as K_∞ for simplicity. Let $K_n = \mathbb{Q}_p(\mu_{p^n})$. For any finite Galois extension L/K_∞, there is $m \geq 0$ and a finite Galois extension L_m/K_m, linearly disjoint from K_∞, such that $L = K_\infty L_m$ (Reference: Serre, *Local fields*, Lemma 6, p. 89). We denote $L_n = K_n L_m$ for $n \geq m$. It is thus sufficient to prove that $\lim_{n\to\infty} v(\mathscr{D}_{L_n/K_n}) = 0$, where $\mathscr{D}_{L_n/K_n}$ is the different. Recall that the different $\mathscr{D}_{L_n/K_n}$ is defined as the inverse as a fractional ideal of the dual of $\mathcal{O}_{L_n}$ under the trace pairing $L_n \times L_n \to K_n$; this follows from the inclusion $\mathscr{D}_{L_n/K_n} \subset \operatorname{tr}(\mathcal{O}_{L_n})\mathcal{O}_{L_n} \subset \mathcal{O}_{L_n}$.

We use higher ramification group, with respect to upper numbering. Recall that upper numbering starts with -1, left continuous, and behaves well with quotients. In particular, for $\Gamma = \mathrm{Gal}(K_\infty/\mathbb{Q}_p)$, $\Gamma^t = \mathrm{Gal}(K_\infty/K_{\lceil t \rceil})$. This can be explicitly computed using local class field theory. Now we have the formula, for E/F,

$$v(\mathscr{D}_{E/F}) = \int_{-1}^{\infty} \left(1 - \frac{1}{|\operatorname{Gal}(E/F)^t|}\right) dt.$$

Thus,

$$v(\mathscr{D}_{L_n/K_n}) = v(\mathscr{D}_{L_n/K_m}) - v(\mathscr{D}_{K_n/K_m}) = \int_{-1}^{\infty} \left(\frac{1}{|\operatorname{Gal}(K_n/K_m)^t|} - \frac{1}{|\operatorname{Gal}(L_n/K_m)|}\right) dt.$$

On the other hand, since $\mathrm{Gal}(L/K_m) \xrightarrow{\sim} \mathrm{Gal}(K_\infty/K_m) \times \mathrm{Gal}(L_m/K_m)$, there is $N > 0$ such that for any $t > N$, $\mathrm{Gal}(L_m/K_m)^t = 1$ and thus $\mathrm{Gal}(L/K_m)^t = \mathrm{Gal}(K_\infty/K_m)^t$, and thus $\mathrm{Gal}(L_n/K_m)^t = \mathrm{Gal}(K_n/K_m)^t$ for any $n > m$. Thus,

$$v(\mathscr{D}_{L_n/K_n}) \leq \int_{-1}^{N} \frac{dt}{|\operatorname{Gal}(K_n/K_m)^t|} \leq \frac{N+1}{|\operatorname{Gal}(K_n/K)^N|},$$

which goes to zero as $N \to \infty$. □

Remark 3.4. (1) Tate showed this for any totally ramified extension for a totally ramified $\mathbb{Z}_p$-extension.

(2) Sen showed that, for K_∞/K a totally ramified extension whose Galois group G is a finite-dimensional p-adic Lie group, then $\overline{\mathbb{Z}}_p/\mathcal{O}_{K_\infty}$ is almost etale. This is also done by comparing upper ramification groups of G with some other group-theoretic filtration of G.

(3) Abhyankar's lemma says that, for a finite Galois tame extension L/K, for any Galois tame extension K'/K such that $|I_{L/K}|$ divides $|I_{K'/K}|$, LK'/K' is unramified. Tate's theorem can be regarded as an almost variant of Abhyankar's lemma.

Another perspective is Scholze's. He proved **almost purity**, which means the following.

Theorem 3.5 (Scholze, almost purity)**.** *Let K be a perfectoid field. For a finite étale extension L/K, $\mathcal{O}_L/\mathcal{O}_K$ is almost étale.*

Definition 3.6. Let K be a complete nonarchimedean field with residue characteristic p. It is caled **perfectoid** if $\Phi : \mathcal{O}_K/p \to \mathcal{O}_K/p$, $x \mapsto x^p$, is surjective, and if the valuation v on K is non-discrete.

Example 3.7. The following are examples of perfectoid fields: $\widehat{\mathbb{Q}_p(\mu_{p^\infty})}$, $\mathbb{C}_p$, $\mathbb{F}_p((t^{1/p^\infty})) := (\mathbb{F}_p[t^{1/p^\infty}]_t^\wedge)[1/t]$.

Proof of almost purity when $\operatorname{char} K = p$. Then K is just perfect. Thus, L is perfect, which means L is also perfectoid. Suppose $\mathcal{O}_K/\operatorname{tr}(\mathcal{O}_L)$ is annihilated by nonzero $\varpi \in \mathfrak{m}_K$. Applying Φ^{-n}, $\varpi^{1/p^n}(\mathcal{O}_K/\operatorname{tr}(\mathcal{O}_L)) = 0$ for all n. Thus, $\mathfrak{m}_K(\mathcal{O}_K/\operatorname{tr}(\mathcal{O}_L)) = 0$. □

To prove almost purity for characteristic zero fields, Scholze introduces the operation called tilting. Given a perfectoid field K of characteristic zero, fix a pseudo-uniformizer $\varpi|p$. Then, $\varprojlim_\Phi \mathcal{O}_K/\varpi$ is of charcateristic p and perfect.

Lemma 3.8. *As topological monoids,*

$$\varprojlim_{x \mapsto x^p} \mathcal{O}_K \xrightarrow{\sim} \varprojlim_\Phi \mathcal{O}_K/\varpi.$$

We denote the lift of $\overline{x} \in \varprojlim_\Phi \mathcal{O}_K/\varpi$ by x.

Proof. For $x = (\cdots, \overline{x}_1, \overline{x}_0)$, choose any lift $x_n \in \mathcal{O}_K$ of $\overline{x}_n$. We claim that $x_i = \lim_{n\to\infty} x_{n+i}^{p^n}$. This is because, for $i \le n$, $x_n^{p^i} (\operatorname{mod} \varpi^{i+1})$ is independent of the choice of x_n (which can be easily proved by induction). $\square$

From Lemma, we have a multiplicative map

$$\varprojlim_\Phi \mathcal{O}_K/\varpi \to \mathcal{O}_K,$$

by sending $\overline{x} \mapsto x^\# := x_0$.

Definition 3.9. The tilt of K is defined to be $K^\flat = \operatorname{Frac}(\varprojlim_\Phi \mathcal{O}_K/\varpi)$.

Another way to define is to define as $K^\flat = (\varprojlim_\Phi \mathcal{O}_K/\varpi)[1/\varpi^\flat]$, where

$$\varpi^\flat = (\cdots, \varpi_2, \varpi_1, 0) \in \varprojlim_\Phi \mathcal{O}_K/\varpi,$$

with $\varpi_1 \in \mathcal{O}_K$ such that $|\varpi_1|^p = |\varpi|$ (this is possible because $v(\mathcal{O}_K)$ is p-divisible), and $\varpi_n^p = \varpi_{n-1} (\operatorname{mod} \varpi)$.

Why is this a field? This is because one can prove that

$$(\varprojlim_\Phi \mathcal{O}_K/\varpi)[1/\varpi^\flat] \cong \varprojlim_{x\mapsto x^p} K,$$

which is an isomorphism of monoids. From this one can also borrow valuation from K, the zeroth entry.

Example 3.10. For $K = \widehat{\mathbb{Q}_p(p^{1/p^\infty})}$, we choose $\varpi = p$. Then, $K^\flat = \mathbb{F}_p((t^{1/p^\infty}))$ with $\varpi^\flat = t$.

Key properties of tilts are as follows.

Lemma 3.11. *We have $|\varpi^\flat| = |\varpi|$, and $\mathcal{O}_{K^\flat}/(\varpi^\flat) \cong \mathcal{O}_K/(\varpi)$.*

Theorem 3.12 (Fontaine–Wintenberger)**.** *Let K be a perfectoid field. There is a bijective correspondence*

$$\left\{ \begin{matrix} \textit{Finite extensions of} \\ K \end{matrix} \right\} \cong \left\{ \begin{matrix} \textit{Finite extensions of} \\ K^\flat \end{matrix} \right\},$$

via tilting. In particular, $G_K \cong G_{K^\flat}$.

This implise that, given a characteristic zero perfectoid field K and a finite extension L of $K^\flat$, there is the **untilt**, $L^\#$, a finite extension of K, which satisfies $(L^\#)^\flat = L$. We explain how to construct $L^\#$. Note that L is the splitting field of a separable polynomial $P(x) = x^d + \sum_{i=0}^{d-1} a_i x^i \in \mathcal{O}_{K^\flat}[x]$. Let

$$P_n(x) = x^d + \sum_{i=0}^{d-1} a_i^{1/p^n} x^i.$$

In particular, $P_n(x)^p = P_{n-1}(x^p)$. If $P(x) = \prod(x - \alpha_i)$, then $P_n(x) = \prod(x - \alpha_i^{1/p^n})$. As $|\alpha_i^{1/p^n} - \alpha_j^{1/p^n}| = |\alpha_i - \alpha_j|^{1/p^n}$, we have $\lim_{n\to\infty} |\alpha_i^{1/p^n} - \alpha_j^{1/p^n}| = 1$, which means $\lim_{n\to\infty} |\operatorname{disc} P_n(x)| = 1$.

Consider

$$Q_n(x) = x^d + \sum_{i=0}^{d-1} (a_i^{1/p^n})^\# x^i \in \mathcal{O}_K[x].$$

We know that

$$Q_n(x) (\operatorname{mod} \varpi) = P_n(x) (\operatorname{mod} \varpi^\flat).$$

Thus,

$$\operatorname{disc} Q_n (\operatorname{mod} \varpi) = \operatorname{disc} P_n (\operatorname{mod} \varpi^\flat).$$

Thus we also know that $\lim_{n\to\infty} |\operatorname{disc} Q_n| = 1$, and that Q_n is separable when n is large enough. Let $N > 0$ be such that, for $n \geq N$, $|\operatorname{disc} Q_n| \geq p^{-\varepsilon}$. Let roots of $Q_n(x)$ be denoted as $\beta_1^{(n)}, \cdots, \beta_d^{(n)}$. As $Q_n(x)^p \equiv Q_{n-1}(x^p) (\operatorname{mod} \varpi)$, we have $Q_{n-1}((\beta_1^{(n)})^p) \equiv 0 (\operatorname{mod} \varpi)$. Since the discriminant of Q_{n-1} is very close to being coprime to p, so we can rearrange roots so that

$$|(\beta_i^{(n)})^p - \beta_i^{(n-1)}| \leq |\varpi| p^\varepsilon.$$

Let $L_n^\#$ be the splitting field of $Q_n(x)$. If n is large enough, we know that

$$|\beta_i^{(n-1)} - (\beta_i^{(n)})^p| < |\beta_i^{(n-1)} - \beta_j^{(n-1)}|,$$

for any $i \neq j$, so by Krasner's lemma, $L_{n-1}^\# \subset L_n^\#$. Thus, for $n \gg 0$, $L_n^\# = L_{n+1}^\# = \cdots$, which we let to be the definition of $L^\#$.

Why do we know $L^\#$ is perfectoid? Note that

$$(\beta_i^{(n)})^p \equiv \beta_i^{(n-1)} (\operatorname{mod} \varpi p^{-\varepsilon}).$$

Although $\mathcal{O}_{L^\#} \supset \mathcal{O}_K[\beta_1^{(n)}, \cdots, \beta_d^{(n)}]$, as the norm of the discriminant goes to 1, we know that

$$\mathcal{O}_{L^\#} \overset{\mathsf{a}}{=} \mathcal{O}_K[\beta_1^{(n)}, \cdots, \beta_d^{(n)}],$$

for any $n \gg 0$. This shows that

$$\Phi : \mathcal{O}_K[\beta_i^{(n)}]/\varpi p^{-\varepsilon} \to \mathcal{O}_K[\beta_i^{(n)}]/\varpi p^{-\varepsilon},$$

is surjective, which means that $L^\#$ is perfectoid.

4. Towards decompletion: Tate's normalized traces

We recall the notation. Let $\mathbb{Q}_p(\mu_{p^\infty}) =: K_\infty$, and let $H_{\mathbb{Q}_p} = \operatorname{Gal}(\overline{\mathbb{Q}}_p/K_\infty)$ and $\Gamma = \operatorname{Gal}(K_\infty/\mathbb{Q}_p)$. Let V be a finite-dimensional representation of $G_{\mathbb{Q}_p}$ over $\mathbb{Q}_p$, and $W = V \otimes_{\mathbb{Q}_p} \mathbb{C}_p$ with diagonal action. There is a business of completion, in that if we let $\widetilde{W}_\infty := W^{H_{\mathbb{Q}_p}}$, it is a vector space over $\widehat{K}_\infty$, and $W^{H_{\mathbb{Q}_p}} \otimes_{\widehat{K}_\infty} \mathbb{C}_p = W$. We would like to remove this completion; we want a K_∞-subspace $W_\infty \subset \widetilde{W}_\infty$ such that

- $\dim_{K_\infty} W_\infty = \dim_{\widehat{K}_\infty} \widetilde{W}_\infty$,
- $W_\infty \otimes_{K_\infty} \widehat{K}_\infty = \widetilde{W}_\infty$,
- W_∞ is Γ-stable.

How do we construct this? The construction can be formulated as follows.

$$W_\infty = \widetilde{W}_\infty^{\Gamma\text{-locally analytic}},$$

where $\widetilde{W}_\infty$ is regarded as a $\mathbb{Q}_p$-Banach space representation of Γ (with a norm given by a choice of lattice in V).

Remark 4.1. Somehow we arrived at analyticity even though we started with a continuous representation. The reason is that because V is finite-dimensional, it is locally analytic. Thus even if we relax finite-dimensionality, we can get a similar theorem for infinite-dimensional yet locally analytic representations.

An important tool is Tate's normalized trace. Let $K_n = \mathbb{Q}_p(\mu_{p^n})$. Note that you can just calculate and discover that $\mathrm{tr}_{K_{n+1}/K_n}(\mathcal{O}_{K_{n+1}}) = p\mathcal{O}_{K_n}$. Thus, one can think of a **normalized trace** as follows.

Definition 4.2 (Tate's normalized trace)**.** For $m \geq n$, we define $\mathrm{tr}_{m,n} = \frac{\mathrm{tr}_{K_m/K_n}}{[K_m:K_n]} : K_m \to K_n$. This is K_n-linear, and $|\,\mathrm{tr}_{m,n}\,| = 1$. It satisfies

$$\mathrm{tr}_{m,n} = \mathrm{tr}_{n+1,n} \circ \mathrm{tr}_{n+2,n+1} \circ \cdots \circ \mathrm{tr}_{m,m-1},$$

and $\mathrm{tr}_{m+1,n}|_{K_m} = \mathrm{tr}_{m,n}$.

From the properties, we can think of passing to the limit, and consider

$$K_\infty = \cup_m K_m \xrightarrow{\cup\,\mathrm{tr}_{m,n}} K_n.$$

Since its norm is 1, we can also pass to the completion and get

$$\mathrm{tr}_n : \widehat{K}_\infty \to K_n.$$

There are some immediate properties.

Proposition 4.3. *(1) For $x \in K_n$, $\mathrm{tr}_n(x) = x$, so in particular $\widehat{K}_\infty = K_n \oplus \ker \mathrm{tr}_n$.*
(2) $|\,\mathrm{tr}_n\,| = 1$.
(3) For $x \in \widehat{K}_\infty$, $\lim_{n\to\infty} \mathrm{tr}_n(x) = x$.
(4) tr_n commutes with the action of Γ.

We introduce further notations. Let $\Gamma_n = \mathrm{Gal}(K_\infty/K_n)$, which is isomorphic to $1 + p^n\mathbb{Z}_p \subset \mathbb{Z}_p^\times \cong \Gamma$. Fix a topological generator γ_n of Γ_n such that $\gamma_n^p = \gamma_{n+1}$. As $K_m^{\gamma_n} = K_n$, $\gamma_n - 1$ is invertible on $\ker \mathrm{tr}_{m,n}$.

Lemma 4.4. *There is $d > 0$ such that, for $x \in K_m$,*

$$|\,\mathrm{tr}_{m,n}(x) - x| \leq d|(\gamma_n - 1)x|.$$

In particular, for $x \in \widehat{K}_\infty$,

$$|\,\mathrm{tr}_n(x) - x| \leq d|(\gamma_n - 1)x|.$$

Proof. The general case follows from an induction on m. For $m = n+1$,

$$\mathrm{tr}_{n+1}(x) - x = \frac{(\gamma_n - 1)x + (\gamma_n^2 - 1)x + \cdots + (\gamma_n^{p-1} - 1)x}{p} = \frac{(p-1) + (p-2)\gamma_n + \cdots + \gamma_n^{p-2}}{p}(\gamma_n - 1)x,$$

so

$$|\,\mathrm{tr}_{n+1}(x) - x| \leq |1/p||(\gamma_n - 1)x|.$$

□

This means that, for $x \in \ker \mathrm{tr}_n$,

$$|x| \leq d|(\gamma_n - 1)x|.$$

Note that $(\gamma_n - 1)$ is invertible on $\ker \mathrm{tr}_n \cap \mathbb{Q}_p(\mu_{p^\infty})$, so $|(\gamma_n - 1)^{-1}| \leq d$ is bounded on it. Since $\ker \mathrm{tr}_n$ is the completion of $\ker \mathrm{tr}_n \cap \mathbb{Q}_p(\mu_{p^\infty})$, this extends continuously to $\ker \mathrm{tr}_n$. We have proved the following.

Proposition 4.5. *The action by* $(\gamma_n - 1)$ *is invertible on* $\ker \mathrm{tr}_n$*, and* $|(\gamma_n - 1)^{-1}| \leq d$*, where* $d > 0$ *is a constant independent of* n*.*

Corollary 4.6. *We have*

$$\dim_{\mathbb{Q}_p} H^i(G_{\mathbb{Q}_p}, \mathbb{C}_p) = \begin{cases} 1 & \text{if } i = 0, 1 \\ 0 & \text{otherwise.} \end{cases}$$

Proof. There is a spectral sequence

$$H^i(\Gamma, H^j(H_{\mathbb{Q}_p}, \mathbb{C}_p)) \Rightarrow H^{i+j}(G_{\mathbb{Q}_p}, \mathbb{C}_p).$$

Since, by almost etale descent $H^j(H_{\mathbb{Q}_p}, \mathbb{C}_p)$ vanishes outside $j = 0$, the spectral sequence degenerates and

$$H^i(G_{\mathbb{Q}_p}, \mathbb{C}_p) = H^i(\Gamma, \widehat{K}_\infty).$$

Choose $n > 1$, and use Γ_n instead;

$$H^i(\Gamma, \widehat{K}_\infty) = H^i(\Gamma_n, \widehat{K}_\infty)^{\Gamma/\Gamma_n} = H^i(\Gamma_n, K_n)^{\Gamma/\Gamma_n} \oplus H^i(\Gamma_n, \ker \mathrm{tr}_n)^{\Gamma/\Gamma_n}.$$

Note that such cohomology can now be computed using a two-term complex given by the map $\gamma_n - 1$. Since $\gamma_n - 1$ acts invertibly on $\ker \mathrm{tr}_n$, $H^i(\Gamma_n, \ker \mathrm{tr}_n) = 0$ for every i. On the other hand, since $\gamma_n = 1$ on K_n, $H^i(\Gamma_n, K_n)$ is K_n for $i = 0, 1$ and 0 otherwise. □

Remark 4.7. We can even talk about an explicit generator of $H^1(G_{\mathbb{Q}_p}, \mathbb{C}_p)$; we have $\log \varepsilon_p : G_{\mathbb{Q}_p} \to \mathbb{Q}_p$, where ε_p is the p-adic cyclotomic character, and consider the unipotent matrices given by this. This encodes an extension of trivial representation by trivial representation.

Remark 4.8. More generally, one can compute $H^i(G_{\mathbb{Q}_p}, \mathbb{C}_p(\chi))$ for $\chi : G_{\mathbb{Q}_p} \to E^\times$, $E/\mathbb{Q}_p$ finite. Namely,

$$\dim_E H^i(G_{\mathbb{Q}_p}, \mathbb{C}_p(\chi)) = \begin{cases} 0 & \text{if } i \geq 2 \\ 1 & \text{if } i = 0, 1, \text{ and } \chi \text{ is smooth (i.e. } \chi(I_{\mathbb{Q}_p}) \text{ is finite)} \\ 0 & \text{if } i = 0, 1 \text{ and } \chi \text{ is not smooth.} \end{cases}$$

Proof. Let $\chi = \chi^{\mathrm{nr}} \chi^r$, where χ^{nr} is unramified and $\chi^r(\mathrm{Frob}) = 1$. Since $\mathbb{Q}_p^{\mathrm{ab}}/\mathbb{Q}_p(\mu_{p^\infty})$ is generated by Frob, we know χ^r factors through Γ. Since χ^{nr} is unramified, it factors through $\mathrm{Gal}(\mathbb{Q}_p^{\mathrm{nr}}/\mathbb{Q}_p) \to E^\times$. By etale descent,

$$\widehat{\mathbb{Q}}_p^{\mathrm{nr}}(\chi^{\mathrm{nr}}) \cong \widehat{\mathbb{Q}}_p^{\mathrm{nr}} \otimes_{\mathbb{Q}_p} E,$$

$G_{\mathbb{Q}_p}$-equivariantly. So, $\mathbb{C}_p(\chi) = \mathbb{C}_p(\chi^r)$, $G_{\mathbb{Q}_p}$-equivariantly. Thus,

$$H^i(G_{\mathbb{Q}_p}, \mathbb{C}_p(\chi)) = H^i(\Gamma, \mathbb{C}_p^{H_{\mathbb{Q}_p}}(\chi^r)) = H^i(\Gamma, \widehat{K}_\infty(\chi^r)).$$

Choose $n \gg 0$ so that $|\chi^r(\gamma_n) - 1|$ is small. We want to use the same decomposition

$$\widehat{K}_\infty(\chi^r) = K_n(\chi^r) \oplus \ker \mathrm{tr}_n(\chi^r),$$

and that at least $\ker \mathrm{tr}_n(\chi^r)$ does not contribute to cohomology.

Lemma 4.9. *Suppose* γ_n *acts continuously on a* $\mathbb{Q}_p$*-Banach space* B *and* $\gamma_n - 1$ *is invertible with* $|\gamma_n - 1| \leq C$*. Then,* $\gamma_n - \alpha$ *is invertible on* B *if* $|\alpha - 1| < C^{-1}$*.*

This is just a consequence of writing $\frac{1}{\gamma-\alpha}$ as a geometric series. Now γ_n acts on χ^r via $\chi^r(\gamma_n)\gamma_n$, so by Lemma, $\gamma_n - 1$ acts invertibly on $\ker \mathrm{tr}_n(\chi^r)$. Similarly for $K_n(\chi^r)$, we have a two-term complex that computes H^i, which gives the desired result. $\square$

Now consider doing the normalized trace for a finite extension $K/\mathbb{Q}_p$. We use the notation $K_\infty = K(\mu_{p^\infty})$ and $K_n = K(\mu_{p^n})$. Then for $n \gg 0$,

$$K_\infty = \mathbb{Q}_p(\mu_{p^\infty}) \otimes_{\mathbb{Q}_p(\mu_{p^n})} K_n,$$

so

$$\widehat{K_\infty} = \widehat{\mathbb{Q}_p(\mu_{p^\infty})} \otimes_{\mathbb{Q}_p(\mu_{p^n})} K_n.$$

Taking $\mathrm{tr}_{K,n} := \mathrm{tr}_n \otimes 1$, we get, for $n \gg 0$,

$$\mathrm{tr}_{K,n} : \widehat{K_\infty} \to K_n.$$

This satisfies similar properties as the original Tate normalized trace:

- is K_n-linear;
- commutes with the action of G_K;
- is continuous, anad $|\,\mathrm{tr}_{K,n}\,|$ is bounded by a constant independent of n (not 1 anymore);
- for $x \in K_n$, $\mathrm{tr}_{K,n}(x) = x$;
- for $x \in \widehat{K}_\infty$, $\lim_{n\to\infty} \mathrm{tr}_{K,n}(x) = x$;
- there is $d > 0$ such that, for $n \gg 0$, $\gamma_n - 1$ is invertible on $\ker \mathrm{tr}_{K,n}$, and $|(\gamma_n - 1)^{-1}| \leq d$.

5. Towards decompletion: locally analytic vectors

We now talk about what we meant by locally analytic vectors. Let G be a d-dimensional p-adic Lie group. Then, there is an open subgroup $G_0 \subset G$ which is pro-p and **uniform**. This means that G_0 is torsion-free and $[G_0, G_0] \subset G_0^p$ if p is odd and $[G_0, G_0] \subset G_0^4$ if $p = 2$. Also, there exist $g_1, \cdots, g_d \in G_0$ such that the map

$$c : \mathbb{Z}_p^{\oplus d} \to G_0, \qquad (x_1, \cdots, x_d) \mapsto g_1^{x_1} \cdots g_d^{x_d},$$

is a homeomorphism. Take $G_n = G_{n-1}^p$, which is a normal open subgroup of G_0, as $G_n = c(p^n \mathbb{Z}_p^{\oplus d})$.

Example 5.1. For $G = \mathrm{GL}_m(\mathbb{Q}_p)$, $G_0 = 1 + p^2 M_m(\mathbb{Z}_p)$ and $G_n = 1 + p^{n+2} M_m(\mathbb{Z}_p)$.

Let E be a $\mathbb{Q}_p$-Banach space. Define $\mathscr{C}(G_0, E)$ to be the space of E-valued continuous functions on G_0. It is a $\mathbb{Q}_p$-Banach space with norm $|f| = \sup_{g\in G_0} |f(g)|$, and has left and right translation actions by G_0.

Definition 5.2. A function $f \in \mathscr{C}(G_0, E)$ is **analytic** (or G_0-analytic) if $f(c(x_1, \cdots, x_d))$ can be written as

$$f(c(x_1, \cdots, x_d)) = \sum_{I=(i_1,\cdots,i_d)\in\mathbb{Z}_{\geq 0}^d} a_I x_1^{i_1} \cdots x_d^{i_d},$$

for $a_I \in E$, and that $\lim_{|I|\to\infty} a_I = 0$. Let $\mathscr{C}^{\mathrm{an}}(G_0, E) \subset \mathscr{C}(G_0, E)$ be the subspace of analytic functions.

This definition of analyticity is independent of choice of $g_1, \cdots, g_d$, and it is thus stable under left and right translation actions.

Remark 5.3. The analytic norm and the continuous norm are in general different; if $f \in \mathscr{C}^{\mathrm{an}}$, only $\|f\|_{G_0} \geq |f|$ holds. An example of strict inequality is when $G_0 = \mathbb{Z}_p$ and $f = x^p - x$; then $\|f\|_{\mathbb{Z}_p} = 1$ and $|f| \leq 1/p$.

Remark 5.4. If $f \in \mathscr{C}^{\mathrm{an}}$ is locally constant, then f is constant.

Definition 5.5. Let W be a $\mathbb{Q}_p$-Banach space representation of G. A vector $v \in W$ is G_0-analytic if the orbit map $o_v : G_0 \to W, g \mapsto g \cdot v$, is analytic, namely $o_v \in \mathscr{C}^{\mathrm{an}}(G_0, W)$. We denote $W^{G_0-\mathrm{an}}$ to be the space of G_0-analytic vectors in W.

It is easy to see that $W^{G_0-\mathrm{an}}$ is a G_0-stable Banach space.

Another point of view is that the evaluation map at the identity gives a map

$$\mathrm{ev}_1 : \mathscr{C}^{\mathrm{an}}(G_0, W) \to W,$$

and if we give $\mathscr{C}^{\mathrm{an}}(G_0, W)$ an action of G_0 by $(g \cdot f)(h) = gf(g^{-1}h)$ (or the G_0-action given by the identification

$$\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p)\widehat{\otimes}_{\mathbb{Q}_p} W =^{\mathrm{an}} (G_0, W),$$

and using the diagonal G_0-action with the left G_0-action on $\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p)$ and the natural G_0-action on W), then this restricts to an isomorphism

$$\mathrm{ev}_1 : \mathscr{C}^{\mathrm{an}}(G_0, W)^{G_0} \xrightarrow{\sim} W^{G_0-\mathrm{an}}, \qquad o_v \mapsto v.$$

Furthermore, the right translation on the function space is identified with the natural G_0-action on $W^{G_0-\mathrm{an}}$.

Definition 5.6. A vector $v \in W$ is **locally analytic** if v is G_n-analyitc for some n. We denote W^{la} to be the space of locally analytic vectors in W. One gives the direct limit topology on W^{la} via

$$W^{\mathrm{la}} = \varprojlim_n W^{G_n-\mathrm{an}},$$

which is a Hausdorff locally Banach spcae. One also gives the G-action as $W^{\mathrm{la}} \subset W$ is G-stable.

Example 5.7. Let $K/\mathbb{Q}_p$ be a finite extension. Let $\Gamma = G = \mathrm{Gal}(K_\infty/K)$ and $\Gamma_n = \mathrm{Gal}(K_\infty/K_n)$. Consider Γ acting on $W = \widehat{K_\infty}$. We claim that

$$\widehat{K_\infty}^{\Gamma_n-\mathrm{an}} = K_n.$$

This is because of the following. As $\mathrm{tr}_{K,m} : \widehat{K_\infty} \to K_m$ is continuous, for $m > n$ and $x \in \widehat{K_\infty}^{\Gamma_n-\mathrm{an}}$, $\mathrm{tr}_{m,n}\, x \in K_m^{\Gamma_n-\mathrm{an}}$. Since the action of Γ_n on K_m factors through Γ_n/Γ_m, this action is locally constant, so $\mathrm{tr}_{K,m}\, x \in K_m^{\Gamma_n-\mathrm{an}} = K_m^{\Gamma_n} = K_n$. As $\lim_{m\to\infty} \mathrm{tr}_{K,m}\, x = x$, $x \in K_n$.

There is a big problem using analyticity, that

$H^i(G_0, \mathscr{C}(G_0, E)) = H^i(G_0, \mathscr{C}^{\mathrm{an}}(G_0, E)) = 0$ for $i > 0$, but $H^i(G_0, \mathscr{C}^{\mathrm{an}}(G_0, E)) \neq 0$ for $i > 0$ in general.

How do we fix this? We assume $G = \mathbb{Z}_p$ from now on. There is Mahler expansion for $f \in \mathscr{C}(\mathbb{Z}_p, E)$;

$$f(x) = \sum_{n=0}^{\infty} a_n \binom{x}{n}, \qquad a_n \in E, \lim_{n\to\infty} a_n = 0.$$

A nice thing is that $\binom{x}{n}$ has norm 1. The Mahler coefficients a_n can be obtained easily by

$$a_n = (\gamma - 1)^n f(0),$$

where γ is the action of $1 \in \mathbb{Z}_p$, namely $\gamma f(x) = f(x+1)$. Actually,

$$(\gamma - 1)f = \sum_n a_{n+1}\binom{x}{n}.$$

Amice has a nice theorem regarding $\mathscr{C}^{\mathrm{la}}$ and $\mathscr{C}^{\mathrm{an}}$.

Theorem 5.8 (Amice). *A function $f \in \mathscr{C}(\mathbb{Z}_p, E)$ is locally analytic if and only if there is $r > 1$ such that $\lim_{n\to\infty} |a_n| r^n = 0$. Also, a function $f \in \mathscr{C}(\mathbb{Z}_p, E)$ is analytic if and only if $\lim_{n\to\infty} \frac{a_n}{n!} = 0$.*

Definition 5.9. For $r > 1$, we define

$$\mathscr{C}^{(r)}(\mathbb{Z}_p, E) = \{f \in \mathscr{C} : \lim_{n\to\infty} |a_n(f)| r^n = 0\}.$$

We endow it with a Banach space structure using the norm $\|f\|^{(r)} = \sup_n |a_n(f)| r^n$. Also, because of the formula for $(\gamma - 1)f$, it is easy to see that $\mathscr{C}^{(r)}$ is stable under translation actions.

Using this, we revisit the cohomology of $\mathscr{C}^{\mathrm{la}}$ and $\mathscr{C}^{\mathrm{an}}$.

Proposition 5.10. *(1) For $r > 1$, $H^1(\mathbb{Z}_p, \mathscr{C}^{(r)}(\mathbb{Z}_p, E)) = 0$.*
(2) We have $H^1(\mathbb{Z}_p, \mathscr{C}^{\mathrm{an}}(\mathbb{Z}_p, E)) \neq 0$.

Proof. We note that $H^1(\mathbb{Z}_p, \mathscr{C}^{(r)}(\mathbb{Z}_p, E))$ is just the γ-coinvariance of $\mathscr{C}^{(r)}(\mathbb{Z}_p, E)$. But this is easily zero, because for any $f(x) = \sum_{n=0}^{\infty} a_n \binom{x}{n}$, $f = (\gamma - 1)g$ for $g(x) = \sum_{n=1}^{\infty} a_{n-1}\binom{x}{n}$. Similarly, for $H^1(\mathbb{Z}_p, \mathscr{C}^{\mathrm{an}}(\mathbb{Z}_p, E))$, whether it vanishes or not is equivalent to asking whether

$$\frac{a_n}{n!} \to 0 \quad \overset{?}{\Rightarrow} \quad \frac{a_n}{(n+1)!} \to 0.$$

This is obviosuly not always true, because $\frac{1}{n+1}$ can be large from time to time. $\square$

Thus we use a better notion of overconvergence.

Definition 5.11. For a $\mathbb{Q}_p$-Banach space representation W of $\mathbb{Z}_p$, let

$$W^{(r)} = \{v \in W : o_v \in \mathscr{C}^{(r)}\}.$$

Equivalently, $v \in W^{(r)}$ if $\lim_{n\to\infty} |(\gamma - 1)^n v| r^n \to 0$.

Then it is easy to show that

$$\mathrm{ev}_0 : \mathscr{C}^{(r)}(\mathbb{Z}_p, W)^{\mathbb{Z}_p} \xrightarrow{\sim} W^{(r)}.$$

Example 5.12. Let W be a unitary Banach space representation of $\mathbb{Z}_p$, namely W° is $\mathbb{Z}_p$-stable. Then $W = W^{(r)}$ for some r (equivalently, by Amice's result, W is locally analytic) if and only if there exists some $n > 0$ such that $(\gamma - 1)^n(W^\circ) \subset pW^\circ$, or equivalently, the image of $\mathbb{Z}_p$ in $\mathrm{End}(W^\circ/pW^\circ)$ is finite.

Suppose $W = W^{(r)} = \mathscr{C}^{(r)}(\mathbb{Z}_p, W)^{\mathbb{Z}_p}$. For $f \in W$, $f(x) = \sum a_n \binom{x}{n}$, we have

$$\|(\gamma - 1)f\| = \sup |a_{n+1}| r^n \leq \frac{1}{r}\|f\|^{(r)}.$$

Thus, for $n \gg 0$, $\|(\gamma - 1)^n f\| \leq \frac{1}{p}\|f\|^{(r)}$. Thus, $(\gamma - 1)^n \mathscr{C}^{(r)}(\mathbb{Z}_p, W)^\circ \subset p\mathscr{C}^{(r)}(\mathbb{Z}_p, W)$, which means $(\gamma - 1)^n(W^\circ) \subset pW^\circ$.

Remark 5.13. In fact, a unitary Banach space representation of a compact p-adic Lie group is locally analytic if and only if it acts via finite quotient on its mod p reduction.

6. Decompletion

Let $K/\mathbb{Q}_p$ be a finite extension, and let V be a $\mathbb{Q}_p$-Banach space representation of G_K. Let $T = V^\circ$, and that $\mathrm{im}(G_K \to \mathrm{Aut}(T/pT))$ is finite.

Remark 6.1. One popular way of finding such representation is to take a family of Galois representation over an affinoid subset of the rigid generic fiber of the deformation ring of a residual Galois representation.

Let $W = V\widehat{\otimes}_{\mathbb{Q}_p}\mathbb{C}_p$. Then, $\widetilde{D} = W^{H_K}$ is a $\mathbb{Q}_p$-Banach space representation of $\Gamma_K = \mathrm{Gal}(K_\infty/K)$, where recall $H_K = G_{K_\infty}$, $K_\infty = K(\mu_{p^\infty})$, $K_n = K(\mu_{p^n})$ and $\Gamma_n = \mathrm{Gal}(K_\infty/K_n)$.

Theorem 6.2 (Decompletion)**.** *In the above context, there is n, which only depends on T/pT, such that*

$$\widetilde{D}^{\Gamma_n-\mathrm{an}}\widehat{\otimes}_{K(\mu_{p^n})}\widehat{K_\infty} \xrightarrow{\sim} \widetilde{D}.$$

Proof. First, one can always replace K by a finite Galois extension L/K by simple Galois descent,

$$W^{H_L,\Gamma_n-\mathrm{an}} = W^{H_K,\Gamma_n-\mathrm{an}} \otimes_{K_n} L_n.$$

In particular one can take the kernel of $G_K \to \mathrm{Aut}(T/pT)$, so we can assume that G_K acts trivially on T/pT.

The theorem is saying roughly that there are enough analytic vectors. The only way to produce analytic vectors is to lift a "mod p analytic vector." We would like to make sense of the notion of Γ_n-analytic vectors in $(T\widehat{\otimes}_{\mathbb{Z}_p}\mathcal{O}_{\mathbb{C}_p})^{H_K}/p$ and lift. This viewpoint is better under our assumption that G_K acts trivially on T/pT, because

$$(T\widehat{\otimes}_{\mathbb{Z}_p}\mathcal{O}_{\mathbb{C}_p})^{H_K}/p \overset{\mathrm{a}}{=} ((T/pT)\otimes\mathcal{O}_{\mathbb{C}_p}/p)^{H_K} = T/pT\otimes(\mathcal{O}_{\mathbb{C}_p}/p)^{H_K} \overset{\mathrm{a}}{=} T/pT\otimes\mathcal{O}_{K_\infty}/p,$$

by almost etale descent, and the action of Γ_K is only via the second factor $\mathcal{O}_{K_\infty}/p$.

Let $E/\mathbb{Q}_p$ be a finite extension, and $\varpi \in E$ be such that $|\varpi| < 1$, and let $r = |\varpi|^{-1}$. We define

$$\mathscr{C}^{(r)}(\mathbb{Z}_p,\mathcal{O}_E) := \{f \in \mathscr{C}^{(r)}(\mathbb{Z}_p,E), |a_n|r^n \le 1\}.$$

This has the decomposition

$$\widehat{\bigoplus_n}\varpi^n\binom{x}{n}\mathcal{O}_E.$$

Now for a p-adically complete $\mathcal{O}_E$-module M with a continuous $\mathcal{O}_E$-linear $\mathbb{Z}_p$-action, we define

$$M^{(r)} := \left(\mathscr{C}^{(r)}(\mathbb{Z}_p,\mathcal{O}_E)\widehat{\otimes}_{\mathcal{O}_E}M\right)^{\mathbb{Z}_p}.$$

Remark 6.3. Now the evaluation map

$$\mathrm{ev}_0 : M^{(r)} \to M,$$

is not injective in general. Indeed, if $f \in \mathscr{C}^{(r)}(\mathbb{Z}_p,M)^{\mathbb{Z}_p}$, then f has a Mahler expansion $\sum_{n=0}^\infty a_n\cdot\varpi^n\binom{x}{n}$. Let $\gamma = -1 \in \mathbb{Z}_p$. Then

$$\gamma\cdot f = \sum\gamma(a_n)\varpi^n\binom{x+1}{n} = \sum(\gamma(a_n)+\varpi\gamma(a_{n+1}))\cdot\varpi^n\binom{x}{n}.$$

So $(\gamma-1)a_n = \gamma(a_{n+1})\varpi$. Thus, if M is p-torsion-free, one can say ev_0 is injective, but in general it is not.

To lift, we need to show that the "obstruction vanishes", which amounts to showing something like vanishing of H^1.

Lemma 6.4. *$H^1(\mathbb{Z}_p, \mathscr{C}^{(r)}(\mathbb{Z}_p, \mathcal{O}_E))$ is ϖ-torsion.*

Proof. Let $f \in \varpi\mathscr{C}^{(r)}(\mathbb{Z}_p, \mathcal{O}_E)$, with $f = \sum b_n \varpi^n \binom{x}{n}$. Letting $g = \sum \frac{b_n}{\varpi} \varpi^{n+1} \binom{x}{n+1}$, we get $(\gamma - 1)g = f$. □

Lemma 6.5. *Let M be as above with the extra assumption that M is p-torsion-free, and see $M[1/p]$ as an E-Banach space. Assume that $(\gamma - 1)$ is invertible on $M[1/p]$ so that $|(\gamma - 1)^{-1}| \leq r$. Then $H^1(\mathbb{Z}_p, \mathscr{C}^{(r)}(\mathbb{Z}_p, M))$ is ϖ-torsion.*

Remark 6.6. In fact, this implies that $M^{(r)} = 0$.

Proof. Let $f \in \varpi\mathscr{C}^{(r)}(\mathbb{Z}_p, M)$, with $f = \sum b_n \varpi^n \binom{x}{n}$. We want $g = \sum a_n \varpi^n \binom{x}{n}$ such that $(\gamma - 1)g = f$, namely $(\gamma - 1)a_n + \varpi\gamma a_{n+1} = b_n$. Since $|(\gamma - 1)^{-1}| \leq r$, as $b_n \in \varpi M$, we have $(\gamma - 1)^{-1} b_n \in M$. So we are led to solve the linear equation

$$a_n + \frac{\varpi\gamma}{\gamma - 1} a_{n+1} = (\gamma - 1)^{-1} b_n,$$

where by the same reason $\frac{\varpi\gamma}{\gamma - 1}$ can be thought as a linear operator from M to M. This system of linear equations is definitely solvable, with explicit formula given by

$$a_k = \sum_{n=0}^{\infty} \left(\frac{\varpi\gamma}{1 - \gamma} \right)^n (\gamma - 1)^{-1} b_{n+k}.$$

□

We would like to apply this to $M = \mathcal{O}_{\widehat{K_\infty}}$.

Lemma 6.7. *There exists $C \in \mathbb{Z}_{\geq 0}$ such that, for all $K/\mathbb{Q}_p$ finite, there exist n and $r > 1$, for some $r^{-1} = |\varpi|$, $\varpi \in K_n$, such that*

$$H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, \mathcal{O}_{\widehat{K_\infty}})),$$

is killed by p^C. Here

$$\mathscr{C}^{(r)}(\Gamma_n, \mathcal{O}_{\widehat{K}_\infty}) := \mathscr{C}^{(r)}(\Gamma_n, \mathcal{O}_{K_n}) \widehat{\otimes}_{\mathcal{O}_{K_n}} \mathcal{O}_{\widehat{K}_\infty}.$$

Consequently,

$$H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, \mathcal{O}_{K_\infty}/p^N)),$$

is also killed by p^C.

Proof. Recall the normalized trace, so that there is $d > 0$ such that for $n \gg 0$, $\gamma_n - 1$ is invertible on $\widehat{K_\infty}/K_n$ with $|(\gamma_n - 1)^{-1}| \leq d$. For $r \geq d$ and $r \in |K_n|$, consider an exact sequence

$$H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, \mathcal{O}_{K_n})) \to H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, \widehat{\mathcal{O}_{K_\infty}})) \to H^1(\Gamma_n, \widehat{\mathcal{O}_{K_\infty}}/\mathcal{O}_{K_n}).$$

The first cohomology is ϖ-torison by Lemma 6.4, and the last cohomology is ϖ-torsion by Lemma 6.5. We can now choose such that $C > 2v_p(\varpi)$. □

Now we go back to the proof of decompletion. Let C, n, r be as in Lemma 6.7 and let $N = C+1$. We can assume further that G_K acts trivially on $T/p^N T$. This is because the maximal pro-p

quotient of G_K is topologically finitely generated (with the number of generators bounded by $H^1(G_K, \mathbb{F}_p)$). Let $M = (T\widehat{\otimes}_{\mathbb{Z}_p}\mathcal{O}_{\mathbb{C}_p})^{H_K}$, $M_s = M/p^s$. Then

$$M_N \stackrel{\mathsf{a}}{=} (T/p^N \otimes \mathcal{O}_{\mathbb{C}_p}/p^N)^{H_K} \stackrel{\mathsf{a}}{=} T/p^N \otimes_{\mathbb{Z}/p^N} (\mathcal{O}_{\mathbb{C}_p}^{H_K}/p^N) \stackrel{\mathsf{a}}{=} T/p^N \otimes \mathcal{O}_{K_\infty}/p^N.$$

Since G_K acts trivially on T/p^N, by Lemma 6.7,

$$H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, M_N)),$$

is p^C-torsion. Consider the diagram

$$\begin{tikzcd}
0 \arrow[r] & M_N \arrow[r, "\times p^s"] \arrow[d, "\times p^C"'] & M_{s+N} \arrow[r] \arrow[d, "\times p^C"'] & M_s \arrow[r] \arrow[d, "\times p^C"'] & 0 \\
0 \arrow[r] & M_N \arrow[r, "\times p^{s-C}"'] & M_{s+N-C} \arrow[r] & M_{s-C} \arrow[r] & 0
\end{tikzcd}$$

Take the Γ_n-r-analytic vectors, and one gets

$$\begin{tikzcd}
M_{s+N}^{(r)} \arrow[r] \arrow[d, "\times p^C"'] & M_s^{(r)} \arrow[r] \arrow[d, "\times p^C"'] & H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, M_N)) \arrow[d, "\times p^C"'] \\
M_{s+1}^{(r)} \arrow[r] & M_{s-C}^{(r)} \arrow[r] & H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, M_N))
\end{tikzcd}$$

Let $N_s = \mathrm{im}(M_s^{(r)} \xrightarrow{\times p^C} M_{s-C}^{(r)})$. That $H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, M_N))$ is p^C-torsion implies that $N_{s+1} \to N_s$ is surjective for all $s > c$. As $\varprojlim_s N_s = \varprojlim_s M_s^{(r)} = M^{(r)}$, it implies that $M^{(r)} \to N_s$ is surjective.

We now study

$$M^{(r)}\widehat{\otimes}_{\mathcal{O}_{K_n}}\widehat{\mathcal{O}_{K_\infty}} \to M.$$

Note that both sides are p-torsionfree, so it is sufficient to study this after taking mod p. Consider the map $M^{(r)}/p \to M_1$; its kernel is easy and is ϖ_n-torsion, where ϖ_n is a uniformizer of K_n. Its image is more mysterious, but this map factors through $N_N = \mathrm{im}(M_N^{(r)} \to M_1^{(r)})$.

We claim that in the diagram

$$\begin{tikzcd}
0 \arrow[r] & (\mathcal{O}_{K_n}/p^N)^{(r)} \arrow[r] \arrow[d] & (\mathcal{O}_{K_\infty}/p^N)^{(r)} \arrow[r] \arrow[d] & (\mathcal{O}_{K_\infty}/\mathcal{O}_{K_n} + p^N)^{(r)} \arrow[d] \\
0 \arrow[r] & \mathcal{O}_{K_n}/p \arrow[r] & \mathcal{O}_{K_\infty}/p \arrow[r] & \mathcal{O}_{K_\infty}/\mathcal{O}_{K_n} + p
\end{tikzcd}$$

the first vertical map is surjective and the third vertical map is zero. As $(\mathcal{O}_{K_n}/p^N)^{(r)} = \mathcal{O}_{K_n}/p^N$, the first vertical map is obviously surjective. To show that the third vertical map is surjective, we show that the image of evaluation $(\mathcal{O}_{K_\infty}/\mathcal{O}_{K_n} + p^N)^{(r)}$ lies in p^C-torsion. Consider

$$0 \to \widehat{\mathcal{O}_{K_\infty}}/\mathcal{O}_{K_n} \xrightarrow{\times p^N} \widehat{\mathcal{O}_{K_\infty}}/\mathcal{O}_{K_n} \to \mathcal{O}_{K_\infty}/\mathcal{O}_{K_n} + p^N \to 0.$$

We claim that $(\widehat{\mathcal{O}_{K_\infty}}/\mathcal{O}_{K_n})^{(r)} = 0$. If $x \in \widehat{\mathcal{O}_{K_\infty}}/\mathcal{O}_{K_n}$, $(\gamma_n - 1)x = y$, then $|x| \le r|y|$. So, $r^k|(\gamma_n - 1)^k x| \ge |x|$, so $x = 0$. Thus $(\widehat{\mathcal{O}_{K_\infty}}/\mathcal{O}_{K_n} + p^N)^{(r)}$ injects into $H^1(\Gamma_n, \mathscr{C}^{(r)}(\Gamma_n, \widehat{\mathcal{O}_{K_\infty}}/\mathcal{O}_{K_n}))$, which is p^C-torsion. Thus this shows that the evaluation map lies in p^C-torsion.

This implies that the image of $(\mathcal{O}_{K_\infty}/p^N)^{(r)} \to \mathcal{O}_{K_\infty}/p$ is the same as $\mathcal{O}_{K_n}/p$. So we know

$$\mathrm{im}(M_N^{(r)} \to M_1) \supset T \otimes \mathcal{O}_{K_n}/p.$$

Thus $M^{(r)}/p \otimes_{\mathcal{O}_{K_n}} \widehat{\mathcal{O}_{K_\infty}} \to M_1$ contains $(T \otimes \mathcal{O}_{K_n}/p) \otimes_{\mathcal{O}_{K_n}} \widehat{\mathcal{O}_{K_\infty}}$, which maps almost surjectively to M_1. Its kernel is ϖ_n^2-torsion, because $\frac{\mathrm{im}(M_N^{(r)} \to M_1)}{T \otimes \mathcal{O}_{K_n}/p}$ is ϖ_n-torsion and $T \otimes \mathcal{O}_{K_n}/p$ and $M^{(r)}/p$ are different up to ϖ_n-torsion. On the other hand, as a map

$$M^{(r)} \widehat{\otimes}_{\mathcal{O}_{K_n}} \widehat{\mathcal{O}_{K_\infty}} \hookrightarrow M,$$

is injective due to p-torsionfreeness. If $n \gg 0$ such that $|\varpi_n^2| > |p|$, then this implies that when you invert p you get isomorphism

$$M^{(r)}[1/p] \widehat{\otimes}_{K_n} \widehat{K_\infty} \cong M[1/p] = \widetilde{D}.$$

After possibly enlarging r so that $r \geq p^{\frac{1}{p-1}}$ (so that by Amice $\mathscr{C}^{(r)} \subset \mathscr{C}^{\mathrm{an}}$), we have $M^{(r)}[1/p] \subset \widetilde{D}^{\Gamma_n - \mathrm{an}}$.

We are thus left to showing $M^{(r)}[1/p] \supset \widetilde{D}^{\Gamma_n - \mathrm{an}}$. Consider the normalized trace

$$t_m : \widetilde{D} = M^{(r)} \widehat{\otimes}_{\mathcal{O}_{K_n}} \widehat{K_\infty} \to M^{(r)} \otimes_{\mathcal{O}_{K_n}} K_m,$$

defined by $1 \otimes \mathrm{tr}_{K,m}$, for $m \geq n$. For $x \in \widetilde{D}^{\Gamma_n - \mathrm{an}}$,

$$t_m(x) \in (M^{(r)} \otimes_{\mathcal{O}_{K_n}} K_m)^{\Gamma_n - \mathrm{an}} = M^{(r)} \otimes_{\mathcal{O}_{K_n}} K_m^{\Gamma_n - \mathrm{an}} = M^{(r)} \otimes_{\mathcal{O}_{K_n}} K_n = M^{(r)}[1/p].$$

Taking $m \to \infty$, we get $x \in M^{(r)}[1/p]$. □

Remark 6.8. Colmez's period ring B_{Sen} is really something like $\mathscr{C}^{(r)}(\Gamma, \mathbb{C}_p)$. Here $D_{\mathrm{Sen}}(V) = \widetilde{D}^{\Gamma - \mathrm{la}}$.

The proof actually shows the following.

Corollary 6.9. *For $m \geq n$,*

$$\widetilde{D}^{\Gamma_n - \mathrm{an}} \otimes_{K_n} K_m = \widetilde{D}^{\Gamma_m - \mathrm{an}}.$$

Note that the almost etale descent

$$W^{H_K} \widehat{\otimes}_{\widehat{K_\infty}} \mathbb{C}_p \cong W,$$

and decompletion

$$\widetilde{D}^{\Gamma_n - \mathrm{an}} \widehat{\otimes}_{K_n} \widetilde{K_\infty} \cong \widetilde{D},$$

are all G_K-equivariant, because it comes from natural inclusions. Thus we have a G_K-equivariant isomorphism

$$\widetilde{D}^{\Gamma_n - \mathrm{an}} \widehat{\otimes}_{K_n} \mathbb{C}_p \cong V \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p,$$

where G_K is thought to act trivially on the left $\mathbb{C}_p$.

We can now define what a Sen operator is. We worked with unitary $\mathbb{Q}_p$-Banach space representation V of G_K with $\mathrm{im}(G_K \to \mathrm{Aut}(T/pT))$ finite, where $K/\mathbb{Q}_p$ is a finite extension. We have, for $W = \mathbb{C}_p \widehat{\otimes}_{\mathbb{Q}_p} V$, with a diagonal G_K-action,

$$W = W^{H_K, \Gamma_n - \mathrm{an}} \widehat{\otimes}_{K_n} \mathbb{C}_p,$$

which is a G_K-equivariant isomorphism, where G_K acts on $W^{H_K, \Gamma_n - \mathrm{an}}$ via $G_K \twoheadrightarrow \Gamma_K$. Since we are taking locally analytic vectors, even $\mathrm{Lie}\, \Gamma_K$ act on $W^{H_K, \Gamma_n - \mathrm{an}}$. Using the cyclotomic character

$\chi : \Gamma_K \to \mathbb{Z}_p^\times$, we have $\operatorname{Lie}\Gamma_K \xrightarrow{\sim} \operatorname{Lie}\mathbb{Z}_p^\times \overset{\log}{=} \mathbb{Q}_p$. The action of $\operatorname{Lie}\Gamma_K$ is continuous, K_n-linear and commutes with the Γ_K-action. By decompletion, we can extend this action $\mathbb{C}_p$-linearly to get an action of $\operatorname{Lie}\Gamma_K$ on W. The action of $1 \in \mathbb{Q}_p = \operatorname{Lie}\Gamma_K$ is called the **Sen operator**, $\theta_V \in \operatorname{End}_{\mathbb{C}_p[G_K]}(W)$.

Remark 6.10. (1) This definition is independent of n, because $W^{H_K,\Gamma_n-\mathrm{an}} = W^{H_K,\Gamma_{n+1}-\mathrm{an}} \otimes_{K_n} K_{n+1}$.

(2) This definition however depends on $\tau : K \hookrightarrow \overline{\mathbb{Q}}_p$ just as Hodge–Tate weights depend on τ.

7. Sample use of Sen theory

If V is finite-dimensional over $\mathbb{Q}_p$, then the condition is immediately satisfied and falls into the realm of classical Sen theory.

Definition 7.1. If V is finite-dimensional, the negatives of eigenvalues of θ_V are called **generalized Hodge–Tate weights** (or Hodge–Tate–Sen weights) of V. The characteristic polynomial of θ_V is called the **Sen polynomial**.

In this terminology, we have the following

Definition 7.2. A finite-dimensional V is **Hodge–Tate** if θ_V is semisimple and the Hodge–Tate–Sen weights are integers. The generalized eigenspace decomposition of $V \otimes_{\mathbb{Q}_p} \mathbb{C}_p$ with respect to θ_V is called the Hodge–Tate decomposition.

Example 7.3. (1) If $V = \mathbb{Q}_p(i)$, corresponding to χ^i, $i \in \mathbb{Z}$,

$$\left(\mathbb{Q}_p(i) \otimes_{\mathbb{Q}_p} \mathbb{C}_p\right)^{H_{\mathbb{Q}_p},\Gamma_n-\mathrm{an}} = \mathbb{Q}_p(i) \otimes_{\mathbb{Q}_p} \mathbb{Q}_p(\mu_{p^n}),$$

and $\theta_{\mathbb{Q}_p(i)} = i$. So V has Hodge–Tate weight $-i$.

More generally, for a Γ_n-analytic character $\psi : \mathbb{Z}_p^\times \to E^\times$ for a finite extension $E/\mathbb{Q}_p$, we consider the Galois representation V given by the character

$$G_{\mathbb{Q}_p} \xrightarrow{\chi} \mathbb{Z}_p^\times \xrightarrow{\psi} E^\times.$$

Then $(V \otimes_{\mathbb{Q}_p} \mathbb{C}_p)^{H_{\mathbb{Q}_p},\Gamma_n-\mathrm{an}} = V \otimes_{\mathbb{Q}_p} \mathbb{Q}_p(\mu_{p^n})$, so $\theta_V = d\psi(1)$, where $d\psi : \mathbb{Q}_p \to E$ is the differential of $\psi : \mathbb{Z}_p^\times \to E^\times$.

(2) Note that $\theta_V = 0$ if and only if the action of $\operatorname{Lie}\Gamma_K$ on $W^{H_K,\Gamma_n-\mathrm{an}}$ is trivial, which means $W^{H_K,\Gamma_n-\mathrm{an}} = W^{H_K,\Gamma_n}$. By Galois descent, $W^{H_K,\Gamma_n} = W^{G_K} \otimes_K K_n$, so $\theta_V = 0$ iff $W = W^{G_K} \widehat{\otimes}_K \mathbb{C}_p$. If V is finite-dimensional, this is equivalent to V being Hodge–Tate of weights $0, 0, \cdots, 0$. We will see later that this is equivalent to V being potentially unramified, or the image of inertia is finite.

This means that θ_V is a scalar given by $i \in \mathbb{Z}$ if and only if

$$V \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p = \left(V(-i) \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p\right)^{G_K} \widehat{\otimes}_K \mathbb{C}_p(i).$$

Thus, for a finite-dimensional V, V is Hodge–Tate if and only if

$$V \otimes_{\mathbb{Q}_p} \mathbb{C}_p = \bigoplus_{i \in \mathbb{Z}} \left(V \otimes_{\mathbb{Q}_p} \mathbb{C}_p(-i)\right)^{G_K} \otimes_K \mathbb{C}_p(i),$$

which is the usual definition. Let $D_{\mathrm{HT}}(V) = \bigoplus_{i \in \mathbb{Z}} \left(V \otimes_{\mathbb{Q}_p} \mathbb{C}_p(-i)\right)^{G_K}$.

(3) Let X be a smooth projective variety over $\mathbb{Q}_p$, and let $V = H^i_{\text{ét}}(X_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_p)$. Then it has a Hodge–Tate decomposition

$$H^i_{\text{ét}}(X_{\overline{\mathbb{Q}}_p}, \mathbb{Q}_p) \otimes_{\mathbb{Q}_p} \mathbb{C}_p \cong \bigoplus_j H^j(X, \Omega^{i-j}) \otimes_{\mathbb{Q}_p} \mathbb{C}_p(j-i),$$

and thus $D_{\mathrm{HT}}(V) = \bigoplus_j H^j(X, \Omega^{i-j})$, "Hodge cohomology." Bhargav Bhatt's lecture tried to say that his theory of prismatic crystals/Higgs crystals give a nice integral structure on $D_{\mathrm{HT}}(V)$ if X has a nice integral model (more correct that the integral lattice coming from coherent cohomology of integral models).

(4) Let $\overline{\rho} : G_{\overline{\mathbb{Q}}_p} \to \mathrm{GL}_d(\mathbb{F}_p)$ be a residual representation, and let R be its universal lifting ring of $\overline{\rho}$, which is a complete local $\mathbb{Z}_p$-algebra with $R/\mathfrak{m}_R = \mathbb{F}_p$. It has a universal deformation $\rho^{\mathrm{univ}} : G_{\mathbb{Q}_p} \to \mathrm{GL}_d(R)$ such that $\rho^{\mathrm{univ}} \bmod \mathfrak{m}_K = \overline{\rho}$. This is universal in the sense that if S is a complete local noetherian $\mathbb{Z}_p$-algebra with $S/\mathfrak{m}_S = \mathbb{F}_p$ and there is $\rho_S : G_{\mathbb{Q}_p} \to \mathrm{GL}_d(S)$ where $\rho_S \bmod \mathfrak{m}_S = \overline{\rho}$ then there is a unique $f : R \to S$ such that $\rho_S = f \circ \rho^{\mathrm{univ}}$. The universal lifting ring has nice properties, such that it has a presentation $R = \mathbb{Z}_p[[X_1, \cdots, X_g]]/(f_1, \cdots, f_r)$.

We would like to use the theory to ρ^{univ}, but $\rho^{\mathrm{univ}} \bmod p$ does not have finite image. It can be remedied as follows. Take

$$R^n = R\left\langle \frac{X_1}{p^{1/n}}, \cdots, \frac{X_g}{p^{1/n}} \right\rangle,$$

which can be thought as the p-adic completion of $R\left[\frac{X_1}{p^{1/n}}, \cdots, \frac{X_g}{p^{1/n}}\right] \subset R[1/p]$. For example, if $R = \mathbb{Z}_p[[x]]$, then $R^n = \mathbb{Z}_p\left\langle X, \frac{X^n}{p}\right\rangle$ is consisted of elements $f(x) = \sum_{i=0}^{\infty} a_i x^i$ where $a_i \in \mathbb{Q}_p$ and $a_i p^{i/n} \to 0$ as $i \to \infty$. The usual $R[1/p] \to \mathbb{C}_p$ should send X to $\mathfrak{m}_{\mathcal{O}_{\mathbb{C}_p}}$ as R is X-adically complete, so $\operatorname{Spm} \mathbb{Z}_p[[X]][1/p]$ is mapped to the open unit disc. On the other hand, $R^n \to \mathbb{C}_p$ should send X to $a \in \mathbb{C}_p$ where $\left|\frac{a^n}{p}\right| \leq 1$, or $|a| \leq |p|^{1/n}$. So, $\operatorname{Spm} R^n[1/p]$ is sent to a closed unit disc of radius $r_n = |p|^{1/n} < 1$.

Now consider

$$\rho^n : G_{\mathbb{Q}_p} \xrightarrow{\rho^{\mathrm{univ}}} \mathrm{GL}_d(R) \to \mathrm{GL}_d(R^n).$$

Now ρ^n satisfies the finite-image-mod-p condition! This is simply because $R/p \to R^n/p$ has finite image, as $X_i^n/p \in R^n$ for every i. Thus we can use the Sen theory on ρ^n, and get the Sen operator $\theta_{\rho^n} \in \mathfrak{gl}_d(R^n \widehat{\otimes}_{\mathbb{Z}_p} \mathbb{C}_p)$. This Sen operator is an interpolation of Sen operators in the following sense: for $x \in \operatorname{Spm} R^n[1/p]$, $R^n[1/p]/\mathfrak{m}_x$ is a finite extension of $\mathbb{Q}_p$, and $\rho^n \bmod \chi : G_{\mathbb{Q}_p} \to \mathrm{GL}_d(k(x))$, $\theta_x \in \mathfrak{gl}_d(k(x) \otimes_{\mathbb{Q}_p} \mathbb{C}_p)$, and $\theta_x = \theta_{\rho^n} \bmod \mathfrak{m}_x$. This is almost immediate by functoriality of the construction of Sen operators.

Now we see why we can't simply work with ρ^{univ}: the "universal Sen operator" θ lies in $\cap_n \mathfrak{gl}_d(R^n \widehat{\otimes}_{\mathbb{Z}_p} \mathbb{C}_p)$, but one can see that this element **does not lie in** $\mathfrak{gl}_d(R \widehat{\otimes}_{\mathbb{Z}_p} \mathbb{C}_p)$! For example, if $\overline{\rho} = 1$ and $d = 1$, then one can check that $R = \mathbb{Z}_p[[X_1, X_2]]$ and $\theta = \log(1 + X_1) = -\sum_n \frac{(-X_1)^n}{n} \in \mathbb{Z}_p[[X]][1/p]$ because of the denominators.

(5) One can use automorphic side to come up with a big Galois representation. For example, Emerton's completed cohomology gives a big Galois representation. Let $K \subset \mathrm{GL}_2(\mathbb{A}_f)$ be an open compact subgroup, and $Y_K = \mathrm{GL}_2(\mathbb{Q}) \backslash \mathbb{H}^{\pm} \times \mathrm{GL}_2(\mathbb{A}_f)/K$. Fix an open compact

subgroup $K^p \subset \mathrm{GL}_2(\mathbb{A}_f^p)$, and let

$$\widetilde{H}^i(\mathbb{Z}/p^n) = \varinjlim_{K_p \subset \mathrm{GL}_2(\mathbb{Q}_p)} H^i(Y_{K^pK_p}, \mathbb{Z}/p^n),$$

which has an action of $\mathrm{GL}_2(\mathbb{Q}_p)$ (by varying p-levels) and $G_\mathbb{Q}$ (by etale cohomology interpretation), and

$$\widetilde{H}^i(\mathbb{Z}_p) = \varprojlim_n \widetilde{H}^i(\mathbb{Z}/p^n),$$

which is p-torsion-free (because Y_K is a curve; it might have p-torsions if Y_K has a higher dimension). This $\widetilde{H}^i := \widetilde{H}^i(\mathbb{Z}_p)[1/p]$ is a Banach space representation of $\mathrm{GL}_2(\mathbb{Q}_p) \times G_\mathbb{Q}$, but we have a similar problem that $\widetilde{H}^i(\mathbb{Z}_p)/p = \widetilde{H}^i(\mathbb{F}_p)$ has an infinite image of $G_{\mathbb{Q}_p}$.

We resolve this as follows. Let $K_p = \Gamma(p^n)$. Then we take $\widetilde{H}^{i,K_p-\mathrm{an}}$.

Lemma 7.4. $\mathrm{im}(G_\mathbb{Q} \to \mathrm{Aut}(\widetilde{H}^{i,K_p-\mathrm{an},\circ}/p))$ *is finite.*

Proof. Since $i = 1$ is the only interesting case, we restrict to that case. Then

$$\widetilde{H}^{1,K_p-\mathrm{an},\circ}/p = (\mathscr{C}^{\mathrm{an}}(K_p)^\circ \widehat{\otimes}_{\mathbb{Z}_p} \widetilde{H}^1)^{K_p}/p \hookrightarrow (\mathscr{C}^{\mathrm{an}}(K_p)^\circ/p \otimes \widetilde{H}^1(\mathbb{Z}_p)/p)^{K_p}.$$

However, $\Gamma(p^{n+1})$ acts trivially on $\mathscr{C}^{\mathrm{an}}(\Gamma(p^n))^\circ/p$; this is analogous to the fact that $p\mathbb{Z}_p$ acts trivially to $\mathscr{C}^{\mathrm{an}}(\mathbb{Z}_p)^\circ/p$, which is simply the assertion that $f(x) \equiv f(x+p) (\mathrm{mod}\, p)$. Thus we can embed further into

$$\hookrightarrow \mathscr{C}^{\mathrm{an}}(K_p)^\circ/p \otimes_{\mathbb{F}_p} \widetilde{H}^1(\mathbb{F}_p)^{\Gamma(p^{n+1})}.$$

Since $\widetilde{H}^1(\mathbb{F}_p)^{\Gamma(p^{n+1})} = H^1(Y_{K^p\Gamma(p^{n+1})}, \mathbb{F}_p)$, this is finite-dimensional, and as $G_\mathbb{Q}$-action comes from this factor, the image of $G_\mathbb{Q}$ is finite! □

Note that this also implies that the Hecke action has finite image mod p, which implies that "Hecke operators are overconvergent."

8. Universal Sen operator

The construction $V \mapsto \theta_V$ is functorial in V, which also respects $\oplus$ and $\otimes$. Thus, by Tannakian formalism, we expect that there is a "universal Sen operator" $\theta \in$"$\mathrm{Lie}\, G_{\mathbb{Q}_p} \otimes_{\mathbb{Q}_p} \mathbb{C}_p$", which we will make precise. Let $L/\mathbb{Q}_p$ be an extension such that $G = \mathrm{Gal}(L/\mathbb{Q}_p)$ is a finite-dimensional p-adic Lie group. Let $G_0 \subset G$ be a uniform pro-p open normal subgroup, corresponding to L/K. Let $V = \mathscr{C}(G, \mathbb{Q}_p)^{G_0-\mathrm{an}}$ be a G-representation with right translation action, and let $T = \mathscr{C}(G, \mathbb{Q}_p)^{G_0-\mathrm{an},\circ}$.

Remark 8.1. From Tannakian point of view, one would maybe want to use $\mathscr{C}(G, bQ_p)$ instead, but then there is an extra assumption that it acts through a finite quotient on the residual representation, so we use analytic vectors. This is justified by the following Lemma.

Lemma 8.2. G_0^p *acts trivially on* T/pT.

We saw its proof above for $G_0 = \mathbb{Z}_p$, and the Lemma holds for more general G.

Now we see V as a representation of $G_{\mathbb{Q}_p} \twoheadrightarrow G$. Then, by Sen theory, there is a Sen operator θ acting on $\mathscr{C}(G, \mathbb{C}_p)^{G_0-\mathrm{an}}$. This is a $\mathbb{C}_p$-linear derivation that commutes with the left invariant action of G. If you think about it carefully, then this exactly means that $\theta \in \mathrm{Lie}\, G \otimes_{\mathbb{Q}_p} \mathbb{C}_p$! Recall

that $\mathrm{Lie}(G)$ is the space of G-left-invariant derivations in G, where $t \in \mathrm{Lie}(G)$ acts on $\mathscr{C}(G, \mathbb{Q}_p)$ via

$$(t \cdot f)(g) = \lim_{n \to \infty} \frac{f(ge^{p^n t}) - f(g)}{p^n}.$$

Definition 8.3. We define $\theta_L \in \mathrm{Lie}\,\mathrm{Gal}(L/\mathbb{Q}_p) \otimes_{\mathbb{Q}_p} \mathbb{C}_p$ to be the Sen operator we just found above. This also lies in $(\mathrm{Lie}\,\mathrm{Gal}(L/\mathbb{Q}_p) \otimes_{\mathbb{Q}_p} \mathbb{C}_p)^{G_{\mathbb{Q}_p}}$.

The quote-unquote universal Sen opeator in "$\mathrm{Lie}\, G_{\mathbb{Q}_p} \otimes_{\mathbb{Q}_p} \mathbb{C}_p$" means the inverse limit of θ_L's,

$$\theta = \text{“}\varprojlim_L \theta_L\text{”} \in \text{“}\varprojlim_L \mathrm{Lie}\,\mathrm{Gal}(L/\mathbb{Q}_p) \otimes_{\mathbb{Q}_p} \mathbb{C}_p\text{”}.$$

This has several properties.

(1) This θ_L is independent of the choice of G_0, because for another choice of $G_0 \geq G_0'$, $\mathscr{C}(G, \mathbb{C}_p)^{G_0-\mathrm{an}} \subset \mathscr{C}(G, \mathbb{C}_p)^{G_0'-\mathrm{an}}$ is dense.

(2) This is universal in the following sense. For a locally analytic Banach representation of G, we have $\mathrm{Lie}\, G$ acting on V, so $\mathrm{Lie}\, G \otimes_{\mathbb{Q}_p} \mathbb{C}_p$ acts $\mathbb{C}_p$-linearly on $V \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p$. Then the Sen operator of V, θ_V, is just the action of θ_L. This is easy to see; if V is, say, G_0-analytic, then V embeds into $\mathscr{C}(G, V)^{G_0-\mathrm{an}}$ via orbit map. Because

$$\mathscr{C}(G, V)^{G_0-\mathrm{an}} = \mathscr{C}(G, \mathbb{Q}_p)^{G_0-\mathrm{an}} \widehat{\otimes}_{\mathbb{Q}_p} V,$$

by functoriality of Sen operator, $\theta_V = (\theta_L \otimes 1)|_{V \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p}$.

(3) The construction of θ_L is functorial in L in the following sense. Let $L_1/L_2/\mathbb{Q}_p$ be such that $G = \mathrm{Gal}(L_1/\mathbb{Q}_p)$, $H = \mathrm{Gal}(L_2/\mathbb{Q}_p)$ are p-adic Lie groups. Then $\theta_{L_1} \equiv \theta_{L_2} (\mathrm{mod}\, \mathrm{Lie}\, H \otimes_{\mathbb{Q}_p} \mathbb{C}_p)$ in $\mathrm{Lie}(G/H) \otimes_{\mathbb{Q}_p} \mathbb{C}_p$.

Example 8.4. (1) For $L = \mathbb{Q}_p^{\mathrm{nr}}$, $\theta_L = 0$. More generally, if L is a finite extension of $\mathbb{Q}_p^{\mathrm{nr}}$ then $\theta_L = 0$.

(2) For $L = \mathbb{Q}_p(\mu_{p^\infty})$, $\mathrm{Lie}\, G = \mathbb{Q}_p$, and $\theta_L \in (\mathrm{Lie}\, G \otimes_{\mathbb{Q}_p} \mathbb{C}_p)^{G_{\mathbb{Q}_p}} = \mathrm{Lie}\, G = \mathbb{Q}_p$ is $1 \in \mathbb{Q}_p$. This characterizes the normalization, and in some sense, this and the above example uniquely characterizes θ_L's.

(3) (Berger–Colmez) Let $L/\mathbb{Q}_p$ have Galois group G. Then, G_0 acts on $\widehat{L}$. What is $\widehat{L}^{G_0-\mathrm{an}}$? Let L/K be the extension corresponding to G_0. Let $V = \mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p)$. Then

$$V \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p = \left(\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p) \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p\right)^{H_K, \Gamma_n - \mathrm{an}} \widehat{\otimes}_{K_n} \mathbb{C}_p,$$

and the action of θ_L on $\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{C}_p) = V \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p$ is sent to the action of $1 \in \mathrm{Lie}\, \Gamma$ on the RHS.

Now take the kernel of θ_L on both sides. Then we get

$$\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{C}_p)^{\theta_L = 0} = (\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p) \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p)^{H_K, \Gamma_n} \widehat{\otimes}_{K_n} \mathbb{C}_p = (\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p) \widehat{\otimes}_{\mathbb{Q}_p} \mathbb{C}_p)^{G_{K_n}} \widehat{\otimes}_{K_n} \mathbb{C}_p.$$

By Galois descent, the RHS is equal to

$$(\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p) \widehat{\otimes}_{\mathbb{Q}_p}) \mathbb{C}_p)^{G_K} \widehat{\otimes}_K \mathbb{C}_p = (\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{Q}_p) \widehat{\otimes}_{\mathbb{Q}_p} \widehat{L})^{G_0} \widehat{\otimes}_K \mathbb{C}_p = \widehat{L}^{G_0-\mathrm{an}} \widehat{\otimes}_K \mathbb{C}_p.$$

Here the G_0-action on the left hand side comes from left translation. Thus, in a sense, if $\dim G = d$, then $\widehat{L}^{G_0-\mathrm{an}}$ is like "$(d-1)$-dimensional". The identification

$$\mathscr{C}^{\mathrm{an}}(G_0, \mathbb{C}_p)^{\theta_L = 0} \cong \widehat{L}^{G_0-\mathrm{an}} \widehat{\otimes}_K \mathbb{C}_p.$$

is explicit, namely $v \otimes 1 \in \widehat{L}^{G_0-\mathrm{an}} \widehat{\otimes}_K \mathbb{C}_p$ corresponds to the (left-translated) orbit map $g \mapsto g^{-1}v$. Thus θ_L is something like the differential equation that all these orbit maps satisfy, which also shows that θ_L is in some sense uniquely characterized. From this, by evaluating at the identity, we get that

$$\theta_L : \widehat{L}^{G_0-\mathrm{an}} \to \mathbb{C}_p,$$

is zero, namely $\theta_L(\widehat{L}^{G-\mathrm{la}}) = 0$. One can say this is a p-adic analogue of Cauchy–Riemann equation.

9. Characterization of $\theta_L = 0$

We would like to discuss the following

Theorem 9.1 (Sen). *$\theta_L = 0$ if and only if $L/\mathbb{Q}_p$ is potentially unramified.*

Here are simple corollaries.

Corollary 9.2. *If $\dim G \geq 2$, then $\theta_L \neq 0$.*

Corollary 9.3. *If $\rho : G_{\mathbb{Q}_p} \to \mathrm{GL}_n(\mathbb{Q}_p)$, then ρ is potentially unramified (i.e. $\rho(I_{\mathbb{Q}_p})$ is finite) if and only if ρ is Hodge–Tate of weights $0, 0, \cdots, 0$ (i.e. $(\rho \otimes \mathbb{C}_p)^{G_{\mathbb{Q}_p}} \otimes_{\mathbb{Q}_p} \mathbb{C}_p = \rho \otimes \mathbb{C}_p$).*

Another way to put is that $\theta_L \neq 0$ if and only if the image of inertia is infinite. By Sen, this is equivalent to L being **deeply ramified** (in the sense of Coates–Greenberg), which means that $\overline{\mathbb{Z}}_p/\mathcal{O}_L$ is almost unramified. By Gabber–Ramero, this is eqiuvalent to $\widehat{L}$ being perfectoid.

Remark 9.4. This is a special case of p-adic monodromy conjecture. There are proofs of p-adic monodromy conjecture avoiding the use of Sen's theorem, so there are some other proofs. But these proofs (with this proof to be presented) are quite difficult somehow.

Sen's proof uses ramification theory and class field theory. Here one can use class field theory because G_n/G_{2n} is abelian! For example, for $G = \mathrm{GL}_d(\mathbb{Z}_p)$, $G_n = 1 + p^{n+2}M_d(\mathbb{Z}_p)$, so $G_n/G_{2n} \cong p^{n+2}M_d(\mathbb{Z}_p)/p^{2n+2}M_d(\mathbb{Z}_p)$, which is abelian as $[p^{n+2}M_d(\mathbb{Z}_p), p^{n+2}M_d(\mathbb{Z}_p)] \subset p^{2n+4}M_d(\mathbb{Z}_p) \subset p^{2n+2}M_d(\mathbb{Z}_p)$. Somehow Sen's proof seems quite roundabout; can't you work on the infinite level?

Corollary 9.5. *Let $I \subset G$ be the inertia group. Then, $\theta_L \in \mathrm{Lie}\, I \otimes_{\mathbb{Q}_p} \mathbb{C}_p$, and furthermore $\mathrm{Lie}\, I$ is the smallest Lie subalgebra $\mathfrak{h} \subset \mathrm{Lie}\, G$ such that $\theta_L \in \mathfrak{h} \otimes_{\mathbb{Q}_p} \mathbb{C}_p$.*

Proof. Let $\mathfrak{h}$ be the smallest such, and let H be a normal subgroup of G with $\mathrm{Lie}\, H = \mathfrak{h}$. Let $L' = L^H$. Then, $\theta_{L'} = 0$, so $L'/\mathbb{Q}_p$ is potentially unramified. $\square$